

String Edit Distance

Problem – We are given 2 strings. We have to find the “edit distance” or the cost of converting one string to other. We are allowed 3 operations – Insert, Delete, Replace. All 3 have equal cost that is 1 unit.

Cost of Insertion :1

Cost of Deletion: 1

Cost of Change : 1

String Edit Distance

		m	o	n	k	e	y
	0	1	2	3	4	5	6
m	1	0	1	2	3	4	5
o	2	1					
n	3	2					
e	4	3					
y	5	4					

String Edit Distance

		m	o	n	k	e	y
	0	1	2	3	4	5	6
m	1	0	1	2	3	4	5
o	2	1	0	1	2	3	4
n	3	2					
e	4	3					
y	5	4					

String Edit Distance

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	0	1	2	3	4	5	6
m	1	0	1	2	3	4	5
o	2	1	0	1	2	3	4
n	3	2	1	0	1	2	3
e	4	3					
y	5	4					

String Edit Distance

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	0	1	2	3	4	5	6
m	1	0	1	2	3	4	5
o	2	1	0	1	2	3	4
n	3	2	1	0	1	2	3
e	4	3	2	1	1	1	2
y	5	4					

String Edit Distance

		m	o	n	k	e	y
	0	1	2	3	4	5	6
m	1	0	1	2	3	4	5
o	2	1	0	1	2	3	4
n	3	2	1	0	1	2	3
e	4	3	2	1	1	1	2
y	5	4	3	2	2	2	1

```
int main ()
```

```
int i, j, l1, l2, m;
```

```
l1 = length();
```

```
l2 = length();
```

```
for (i = 0; i <= l1; i++)
```

```
    t[i][0] = i;
```

```
for (i = 1; i <= l2; i++)
```

```
    t[0][i] = i;
```

```
for (i = 1; i <= l1; i++)
```

```
    for (j = 1; j <= l2; j++)
```

```
    {
```

```
        m = min(t[i-1][j], t[i][j-1]) + 1;
```

```
        t[i][j] = min(m, t[i-1][j-1] + (word1[i-1] == word2[j-1] ? 0 : 1));
```

```
    }
```

Minimum Edit Distance Example

String Edit Distance

Problem – We are given 2 strings. We have to find the “edit distance” or the cost of converting one string to other. We are allowed 3 operations – Insert, Delete, Replace.

Cost of Insertion :1

Cost of Deletion: 1

Cost of Change : 2

EXAMPLE 2

		$j \rightarrow$				
		0	B	A	B	B
		0	1	2	3	4
A	0	0	1	2	3	4
	1	1	2	1	2	3
	2	2	3	2	3	4
	3	3	2	3	2	3
	4	4	3	2	3	4
B	5	5	4	3	2	3